P1-09 — Claim-by-Claim Overlap Matrix

Date: 21 October 2026 **Companion:** P1_09_GATE_G1_NOVELTY_AUDIT.md (§2–§4 give the reasoning; this file is the matrix + per-source record). **Rules:** repo labels L1–L10 as in 01_LITERATURE_AUDIT.md / P1_01_FULL_TEXT_THEOREM_EXTRACTION.md §0. "Verified" = seen in lawfully accessed text (full text or abstract/metadata, marked). "In inspected text" for absences refers only to the texts actually read (arXiv full texts for L4/L8; abstracts/publisher pages for the rest). Inaccessibility is never treated as absence.

Legend: ● = present (verified), ◐ = partial/related or abstract-level only, ? = unresolved, — = not present in the inspected text, n/a = not applicable.

Our claims (corrected blueprint, frozen 21 Oct 2026, commit 1c67fda blueprint content):

- **K1** Hull generator + dimension at exponent level, repeated-root (n = Pn_0, p | n allowed), compatible twists (T3/T4).
- **K2** Joint (dim_{F_q} C, dim_{F_q} Hull_k) enumerator over **labeled** assignments over A = ∏ K_s (T6).
- **K3** Per-orbit transfer-matrix trace formula (T5).
- **K4** LCD count 2^{#orbits} (C3); self-orthogonal/dual-containing criteria + closed-walk counts (C4).
- **K5** Mean formulas: corrected hull-mean split (C5 as corrected by PC-1) and code-dimension mean.
- **K6** Variance closed forms in P for L = 1, 2, ≥ 3 (C6 as integrated from P1-06).
- **K7** P = 1 recovery of the simple-root formulas (Gate-G3 identity).

Claim | L1 (finite-field, 2023) | L2 (average dims) | L3 (small hull dims) | L4 (affine, arXiv full text) | L8 (multi-twisted, arXiv full text) | L9/L10 & Check-D neighbours

K1 hull generator/dimension, repeated-root | ? (abstract: hull-dimension formula exists; scope/hypotheses unseen) | — (averages, not per-code formula) | ◐ (conditions for dim 1, 2 incl. `p | n` cases; not the general exponent formula) | ◐/● (hull generator + dimension formula over A; repeated-root Thm. under extra ord λ hypotheses, simple-root idempotent framework) | ◐ (GPM hull dimension formulas, twisted setting; not our affine exponent lattice) | ◐ (repeated-root hull/LCD literature at publisher-scope level)

K2 joint enumerator, labeled, affine product | ? (abstract: "count of codes for given hull dimension" — joint vs marginal unseen) | — | — | — (counts of attainable dimension ∗values∗; code enumeration over A left open in its conclusion) | — (not in inspected text) | — (joint ∗weight∗ enumerators of LCD pairs etc. are different objects)

K3 transfer-matrix trace per orbit | ? | — | — | — | — | —

K4 LCD / self-orth. counts & criteria | ? | — | ◐ (small-dim existence, not counts) | ◐ (LCD **condition**; not our labeled count 2^{#orbits}) | ◐ (LCD-type results in GPM setting) | ◐ (LCD enumerations in other models, e.g. serial rings)

K5 means | ? (no mean in abstract) | ● (average Galois hull dimensions over finite fields / R_{m,q} — abstract verified; different model, mean as headline there) | — | — | — | ◐ (classical hull-average literature)

K6 variance in P | ? | ? (variance not verified) | — | — | — | —

K7 P = 1 recovery | n/a | n/a | n/a | n/a (their setting is simple-root; our P=1 case is a specialization, not their theorem) | n/a | n/a

Per-source record

Repo L1 — Debnath, Prakash & Islam, "Galois hulls of constacyclic codes over finite fields", Cryptogr. Commun. 15(1), 111–127 (2023). DOI 10.1007/s12095-022-00591-6; correction DOI 10.1007/s12095-022-00602-6, 15, 129–130. zbMATH Zbl 1518.94136 / correction Zbl 1518.94137. OpenAlex W4283360609; Semantic Scholar CorpusId 250008629.

- **Verified (abstract/metadata level only):** the paper "presents a formula for the dimension of Galois hulls of constacyclic codes" (Springer abstract snippet, lawful page, re-read 21 Oct 2026) and, per its abstract as recorded in P1-01 §5.1 and corroborated by the introduction of repo L4 (ScienceDirect snippet: "They also counted the number of constacyclic codes of a given Galois hull dimension"), a count of codes by hull dimension. Authors/affiliation (IIT Patna), pages, dates

verified via OpenAlex/Semantic Scholar (fresh 21 Oct 2026 responses: is_oa = false, any_repository_has_fulltext = false, openAccessPdf empty).

- **Not verified:** every theorem statement and hypothesis — in particular: whether p | n (repeated roots) is included or excluded; whether the count is of labeled codes, of codes with a given hull dimension only (marginal), or joint in (dim C, dim Hull); the orbit/factor bookkeeping (cf. B_i sets in the correction); restrictions on q, k, λ.
- **Shared vs different:** potentially shares K1's finite-field specialization and some counting territory (K2/K4 territory cannot be excluded). Even in the worst case for us, L1 is finite-field only (title/abstract), so K2's affine-product assembly, K3, K5's corrected split in P, and K6 have no counterpart ∗in the verified record∗ — but that record is incomplete.
- **Status: UNRESOLVED.** Exact comparison impossible until a lawful full text is read. This single source blocks Gate G1 closure.

Repo L2 — "Average dimensions of Galois hulls of constacyclic codes", Adv. Math. Commun. DOI 10.3934/amc.2025010; Zbl 1579.94062.

- **Verified (abstract):** formulas for average dimensions of Galois hulls of constacyclic codes over finite fields (and, per the paper's scope as recorded in P1-01 §6.3, related ring cases).
- **Shared/different:** shares the ∗topic∗ of hull means (K5) but in the finite-field (and R_{m,q}) model with the mean as headline; our C5 is a corollary of the joint enumerator in the repeated-root affine labeled model, with the L = 1/L ≥ 2 split. No joint enumerator, transfer trace, or P-variance verified.
- **Status: PARTIAL (abstract-level).** Consequence already locked (GAP-12): the mean must never be our headline novelty.

Repo L3 — "Study of small Galois hull dimensions of constacyclic codes", Adv. Math. Commun. DOI 10.3934/amc.2025054.

- **Verified (publisher page/table):** conditions for Galois hull dimensions 1 and 2, including cases with p | n (Table 1 conditions, e.g. ord(α) | (1 + p^{e−ℓ}), α^n = λ; fetched snippet 21 Oct 2026 records "Thm. 3.9" **as displayed on that publisher page**).
- **Shared/different:** overlaps K1 only in the small-dimension corner (dim Hull ∈ {1, 2} existence/conditions). No enumerator, counts of all codes, transfer traces, or moments.
- **Status: PARTIAL (publisher-page level).** Consequence: we may not claim to be first to treat repeated-root hulls or small hull dimensions.

Repo L4 (command "L1") — Debnath, Islam, Martínez-Moro & Prakash, "Galois hulls of constacyclic codes over affine algebra rings", Discrete Math. 349(2) (2026), Art. 114750. DOI 10.1016/j.disc.2025.114750; arXiv 2412.08512 (full text inspected in P1-01); Zbl 1582.94185.

- **Verified (arXiv full text; P1-01 §3):** square-free affine algebra A; candidate-first pairing convention (differs from ours — convention is not contribution); hull via component idempotent generators Σ e_s·lcm(g, h^#)-type formula; hull-dimension formula; a repeated-root extension theorem (arXiv Thm. 8 in P1-01's extraction) under additional hypotheses (gcd(ord λ, n′) = 1, ord λ | 1 + p^{e−k}-type conditions); hull generator Σ e_{S(ν)} lcm(g, h^#) over components (Thm. 7, arXiv numbering), Galois dual generator (Thm. 5); counts of **attainable hull-dimension values** (Cor. 3.5), not of codes; its conclusion (§5) leaves enumeration of codes over A open.
- **Shared vs different:** shares the affine setting and hull-generator/dimension territory (K1 partially — under different hypotheses and a semisimple/idempotent method that does not apply to our non-reduced quotient K_s[x]/(x^n − λ_s) when P > 1). Does **not** contain (in inspected text): our exponent-lattice parametrization of ∗all∗ codes for P > 1 with the transfer trace (K2/K3), labeled joint enumerator, C3/C4 counts, K5/K6 moments.
- **Status: VERIFIED at arXiv full-text level; publisher-version equivalence unverified**

(metadata/snippet consistency only). Consequence: K1 alone cannot be sold as the novelty; the enumerator/transfer/moments must carry the claim, with L4 cited and contrasted.

Repo L8 (command “L4”) — Ramy Abu Eldin & Patrick Solé, “Generator polynomial matrices of the Galois hulls of multi-twisted codes”, Finite Fields Appl. 109 (2026), Art. 102712. DOI 10.1016/j.ffa.2025.102712; arXiv 2305.15237 (full text inspected in P1-01 §4); Zbl 1576.94064.

- **Verified (arXiv full text, arXiv numbering):** Galois hull machinery for multi-twisted (GPM) codes — hull-dimension via generator-matrix rank (Thm. 8), Galois dual twist type (Thm. 9), Euclidean-dual GPM relation (Thm. 10), hull type under Λ^{−p^{−κ}} = Λ (Cor. 11), determinantal-divisor hull dimension (Thm. 25), GPM for the hull (Thm. 28), LCD/self-orthogonal criteria (Cors. 26/29); candidate-first pairing; constacyclic codes are the index-1 case and no general gcd(m_j, p) = 1 restriction was found in the inspected text. Publisher numbering demonstrably differs from arXiv (P1-02 §2.2), so only arXiv numbering is ever quoted here.
- **Shared vs different:** shares hull-dimension technology in a different code family; no joint (dim, hull) enumerator over an affine exponent lattice, no per-orbit transfer trace, no P-moments in the inspected text.
- **Status: VERIFIED at arXiv full-text level (publisher equivalence unverified).**

Repo L9/L10 and Check-D neighbours (publisher-scope / abstract level; P1-01 §6.4, P1-02 §3)

- Repeated-root constacyclic hull/LCD literature exists (publisher-scope records), so “repeated-root hulls are untouched” would be false. Classical hull enumeration/averages (Sendrier–Skersys tradition), joint ∗weight∗ enumerator of an LCD code and its dual (arXiv:1511.08889), DC/FC hull-dimension enumeration (arXiv:2307.05449), and hull enumeration over serial rings are **different objects** from K2/K3; none, in the inspected records, gives the labeled affine repeated-root joint (dim C, dim Hull) transfer-trace enumerator.
- **Status: PARTIAL (scope-level).** Databases searched (P1-02 Check D, re-confirmed with a fresh targeted search 21 Oct 2026 that returned no counterpart): arXiv, zbMATH Open, OpenAlex, Semantic Scholar, publisher pages. **Not searched: MathSciNet, Scopus, Web of Science** (no lawful workspace access). Coverage is non-exhaustive; non-discovery is not novelty.

Matrix conclusion

No inspected source contains K2+K3 (joint labeled affine repeated-root enumerator by transfer traces) or K6 (variance in P), and K5 is headline-owned by L2 in its own model. K1 is substantially pre-occupied (L4 verified; L1 unresolved). The defensible contribution centre is therefore **K2 + K3 + K6 (with K4/K5 as corollaries)** — subject to the L1 caveat, which is exactly why Gate G1 cannot yet close (see the companion audit).